Project Proposal: Approximating Total Variation Distance

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May 6, 2026

Based on: A simple polynomial-time approximation algorithm for the total variation distance between two product distributions

1 Paper Overview

This paper technically bypasses the computational challenge of calculating the **Total Variation (TV) distance** between two product distributions P and Q which was proven to be **#P-hard**. While other papers that the authors mention give polynomial-time randomised approximation algorithms, they tend to work only in specific, restricted cases (such as when one distribution has marginals over 1/2 and dominates the other).

The authors provide the first general **polynomial-time approximation algorithm** based on the Monte Carlo method that works on any two product distributions. Significantly, their method overcomes the limitations of "naive" Monte Carlo sampling—which fails when the TV distance is exponentially small—by using a **coordinate-wise greedy coupling** as a proxy for the optimal coupling as it is difficult to compute. Instead, each coordinate is coupled optimally and independently which is then used to condition a distribution on the event that two samples are different. Their goal is to boost the probability that two distributions are different, overcoming the limitation of a small TV distance. By sampling from this specialized conditional distribution π, the algorithm can provide an (ϵ, δ)-approximation in $O(n^2/\epsilon^2 \log(1/\delta))$ time. The authors claim that their expected estimate - the ratio between the probabilities of the assignment in the optimal coupling and the greedy coupling- is lower bounded by $\frac{1}{n}$ where n is the number of coordinates in a distribution.

2 Proposed Research Directions

2.1 Pursuit of a Deterministic Approximation Algorithm

As previously stated, there have been developments in deterministic approximation algorithms for the TV distance of specific restricted types of product distributions. While the authors describe their randomized algorithm for general product distributions, they explicitly state that a major open question remaining is whether **deterministic** approximation algorithm exists for the TV distance of any product distribution. They suggest a positive answer because of the mathematical connections between the counting knapsack problem and calculating exact TV distance.

- **Goal:** Investigate the connection between TV distance and **counting knapsack solutions**, for which deterministic schemes already exist.
- **Approach:** We aim to explore if the greedy coupling framework can be adapted into a deterministic branching program or dynamic programming approach to remove the reliance on random sampling.

2.2 Generalizing to Continuous Distributions

The paper focuses on finite sets $[q]^n$, so a valuable extension to explore is whether the greedy coupling technique can be generalized to **product distributions over continuous domains**. If the per-coordinate $d_{TV}(P_i, Q_i)$ can be computed or approximated for continuous marginals, the overall framework might still hold. This could help eliminate the computational bottleneck of high-dimensional integration used for real-world datasets that rely on continuous product distributions.

- **Goal:** Adapt the **Monte Carlo-based approximation algorithm** to handle product distributions over continuous state spaces.
- **Approach:** We will determine if the **conditional distribution** π and the **estimator** $f(\omega)$ can be reformulated using probability density functions instead of discrete mass functions with 3 components: translating the discrete probability sums to integrals, adapting to a continuous sampling design, and analyzing the variance and expectation.

2.3 Implementing Parameter Independent Adaptive Sampling

The current algorithm dictates a strict, pre-calculated number of Monte Carlo samples based entirely on worst-case theoretical boundaries. However, in practice, **the actual variance for many distributions is often much lower than these strict mathematical worst-case scenarios**. This means the algorithm frequently runs much longer than necessary. By modifying how the algorithm determines when it has enough data, we can drastically reduce the constant runtime factors for real-world, practical applications.

- **Goal:** Improve the practical runtime of the algorithm by replacing the fixed sample size with a dynamic, "parameter-free" implementation.
- **Approach:** We aim to introduce an **adaptive Monte Carlo stopping rule**. Instead of forcing the algorithm to run a predetermined number of times, we will modify it to calculate the empirical variance on the fly as it samples. This will allow the simulation to safely stop early as soon as a target statistical confidence threshold is met.